Decision Support Systems 194 (2025) 114455

Contents lists available at ScienceDirect

Decision Support Systems

journal homepage: www.elsevier.com/locate/dss

Leveraging meta-path and co-attention to model consumer preference stability in fashion recommendations

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ARTICLE INFO

Keywords:
Outfit recommendations
Preference stability
Attention mechanism
Metapath

ABSTRACT

With countless outfit combinations available, consumers often experience choice overload. Two key challenges that significantly impact the quality of recommendation systems are recommendation accuracy and fluctuations in consumer preferences. Previous works primarily extracted generic product features and modeled the compatibility of fashion items, overlooking the relationships hidden in user-product interactions and the evolution of consumer preferences. Unfortunately, this evolution of consumer preferences has not received much attention in the RS studies. To address these limitations, we propose a GPA-BPR (General compatibility and Personalized preference with co-Attention mechanism) framework, which integrates multimodal insights for practical outfit evaluation and utilizes item-user-item meta-paths to capture consumers' stable preferences. Experiments demonstrate significant performance improvements. The co-attention mechanism in our framework effectively enhances recommendations based on meta-path contexts compared to similar previous studies.

1. Introduction

Outfit matching poses unique challenges in e-commerce. Consumer preferences in fashion can change due to evaluation stages, trends, and the symbolic values of products [1]. This shift occurs through aesthetic perceptions and learning, where consumers form an initial impression of an outfit's aesthetics, explore new possibilities, and refine their selection based on new options [1]. Another challenge is the large number of top-and-bottom combinations, leading to choice overload – a phenomenon when the complexity of a decision exceeds an individual's cognitive capacity [2]. A key reason for choice overload is feature complementarity, which refers to the extent to which product features complement each other to meet consumer needs [2]. In outfit selection, multiple tops may pair well with a single bottom, creating an extensive range of possible combinations. Choice overload can lead to consumer dissatisfaction, reduced confidence in decision-making, and post-decision regret [2,3]. Well-crafted recommender systems (RSs) can help alleviate choice overload [4] by reducing the search cost and simplifying product comparisons.

Despite research in automated outfit recommendations, issues in methodology, data representation, and theoretical assumptions hinder the effectiveness of current approaches to reduce the choice overload problem. First, many prior fashion recommendation studies [5,6] have utilized Bayesian Personalized Ranking (BPR), a pairwise ranking optimization method widely used in recommendation tasks. However, while BPR is effective for pairwise ranking, it may overlook finer details or specific attributes within the data. The present study introduces meta-path and co-attention into the outfit recommendation algorithm to address this limitation. More specifically, it allows the model to distinguish user preferences through unique paths by recognizing changes in user preferences as well as styles. Second, most outfit-matching studies recommend outfit pairs based on the complementary fit from visual or textual product descriptions [7–10]. The fundamental assumption is that such insights compiled from other consumers will be relevant to each individual, but this assumption is not always valid and could lack a personal touch [1]. Furthermore, user preference is not a stable concept. It shifts towards “a state of coherence” before the final choice [11]. Consumers usually start by constructing their initial preferences and gradually stabilizing them towards a goal [12], which means that a consumer's preference may not converge to a stable preference at the time when they first use an RS. During this preference construction time, preferences are context-sensitive, and preference calculation is still

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https://doi.org/10.1016/j.dss.2025.114455
Received 17 April 2024; Received in revised form 8 April 2025; Accepted 18 April 2025
Available online 20 April 2025
0167-9236/

Table 1
Key related works on personalized fashion recommendation.

Authors	Year	Data source	Method					Modality			TC	Contribution	Limitation
			DL	BPR	AM	MP	G	T	V	O			
He & McAuley [13]	2016	Amazon		V					V			Uncovers both visual and latent factors in fashion.	Simplistic image feature extraction.
Kang et al. [16]	2017	FashionVC		V					V			Uses GAN to generate images tailored to user preferences.	Does not consider item compatibility.
Yu et al. [15]	2021	Amazon		V					V	V	V	Enrich samples by considering item similarity in the aesthetic space.	Ignoring higher-order connections and not exploring aesthetic features in explicit feedback.
Tay et al. [25]	2018	DeepFashion			V					V		Co-attention mechanism captures personalized preferences from multiple user reviews.	Relies heavily on the quality and relevance of reviews.
Song et al. [7]	2017	Polyvore		V				V	V			Models visual and contextual modalities with BPR to capture user preferences.	Lacks consideration of other modalities and deeper interactions.
Song et al. [17]	2018	Amazon		V	V			V	V			Integrating fashion domain knowledge into an attentive knowledge distillation model.	Matching rules are difficult to define without domain experts.
Song et al. [14]	2018	iQon3000		V				V	V			Incorporating general compatibility modeling for item–item interactions and personal preference modeling for user–item interactions.	Fuse the two components of modeling linearly.
Zhou et al. [24]	2018	FashionVC			V			V	V			Attention mechanism categorizes user behaviors from diverse data.	Concatenating behavior sequences fails to capture interaction effects between behaviors.
Yang et al. [18]	2019	iQon3000					V	V	V			Model category-based compatibility using complementary relations.	Uses only broad categories, missing finer attributes.
Han et al. [31]	2019	FashionVC Polyvore	V						V		V	Bi-LSTM predicts item compatibility sequentially.	Over-reliance on recent clothing information overlooks long-term compatibility.
Laenen et al. [23]	2020	Polyvore			V			V	V			Attention-based methods enhance item representations.	Focuses on vision-text fusion without incorporating other modalities.
Ding et al. [29]	2021	Amazon					V			V	V	Sequential modeling captures long- and short-term user preferences.	Infers short-term preferences from other users, potentially missing individual needs.
Song et al. [25]	2021	Polyvore					V	V	V	V		Graph Neural Networks model intra-item and cross-modal compatibility.	Focuses on item relationships, under-emphasizing personalization.
Guan et al. [22]	2022	iQon3000			V	V		V	V	V		Captures complex user-item relationships using heterogeneous graphs and metapaths.	Focuses on item relationships, neglecting dynamic user preferences and behavior patterns.
Jing et al. [20]	2023	FashionVC					V	V	V	V		Integration multi-modal data, utilize category relationships in items, and captures high-order correlations.	Personalization aspects and rich extra-connectivity information associated with items are not fully explored.
Xu et al. [21]	2024	Polyvore			V		V	V	V			Considering two granularities of the category modality space.	Neglects incorporating users' personal preferences.

Note: DL, traditional deep learning only; BPR, Bayesian personalized ranking; AM, attention mechanism; MP, metapath-based HIN; G, graph-based HIN; T, textual description including category; V, visual; O, other attributes; TC, temporal consideration.

being performed [12]. Most RS studies have no discretion between preference construction and stabilization phases. As a result, the true consumer preference in the stabilization stage is often obscured by consumer explorations during the preference construction stage. Third, a multi-modality view of products and user preferences is a new way of moving RS research forward. The previous two issues highlight that recommendations based solely on others' experiences may be less relevant to individual consumers. Existing findings that fail to distinguish the preference stages may have limited practical applicability. Addressing these challenges requires methodological and data representation changes in recommendation studies, meaning previously optimized models and parameters may no longer be effective. Additionally, existing methods often model user-item interactions or item-item matching patterns separately, neglecting the mutual influence between meta-paths and the associated user-item pairs. This further hinders optimization in recommendations. Therefore, this study aims to address these two problems.

To address the aforementioned challenges, we frame our research questions (RQs) as follows: RQ1 asks how our extended BPR method, which incorporates meta-path and co-attention mechanisms, compares with existing approaches. RQ2 explores how to model stable consumer preference. Lastly, RQ3 examines how the proposed framework generalizes across various combinations of personal preferences and product features.

In this study, we propose a personalized outfit modeling framework using GPA-BPR (General compatibility and personalized preference with co-attention mechanism), which assesses outfit compatibility using both multi-modal data (textual descriptions and images) and individual preferences. GPA-BPR consists of two core elements: general outfit-matching and personalized preference modeling. The former learns a shared latent space for textual and visual data, capturing item-item interactions for outfit matching. The latter uses meta-path and co-attention to effectively extract implicit feedback and comprehensively capture multi-relational connections between users and items. Together, they capture item compatibility and rich personalized information.

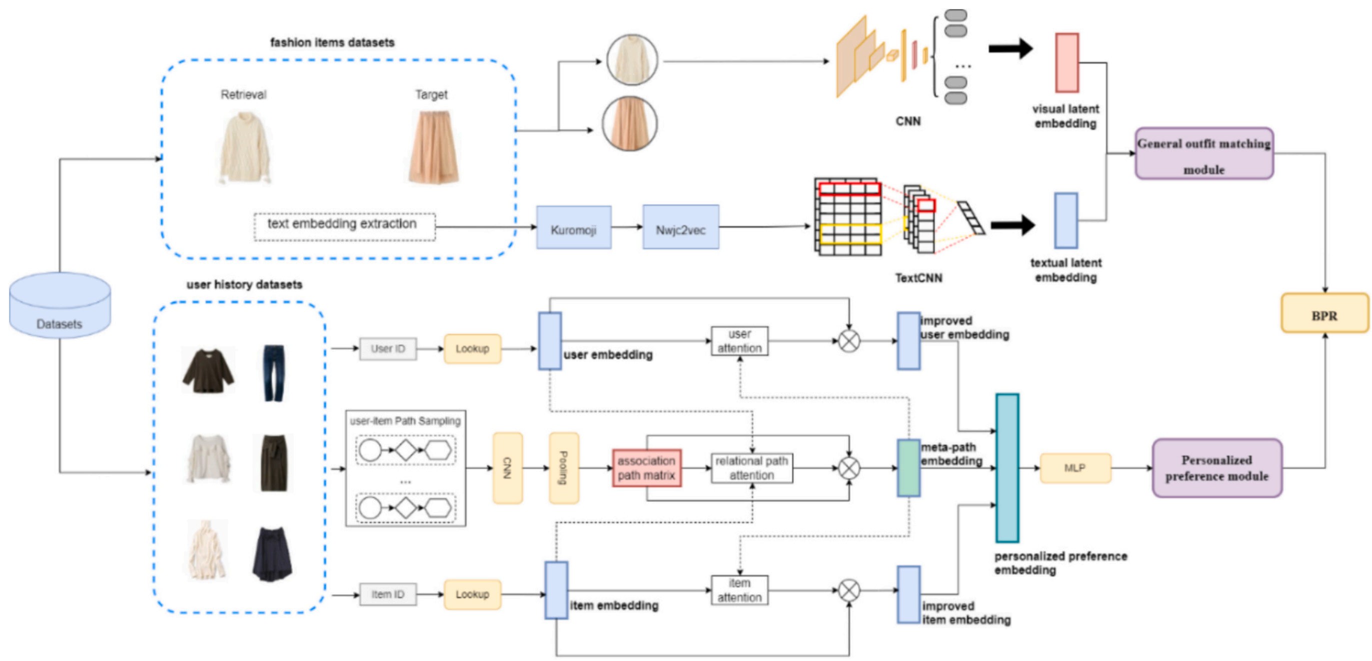

Fig. 1. Overview of the proposed GPA-BPR framework.

2. Related work

2.1. Current methodologies for outfit recommendation

Table 1 summarizes key fashion recommendation studies. Most personalized fashion recommendation studies [13–17] have used Bayesian Personalized Ranking (BPR) [6], which enhances personalized rankings based on implicit user feedback. He and McAuley [13] introduced a Visual BPR (VBPR) model, integrating matrix factorization (MF) to address data sparsity, but its simplistic feature extraction overlooks critical visual information. Song et al. [7] introduced a dual autoencoder for item modalities and preferences but lacked personalization. Kang et al. [16] proposed Deep Visual BPR with pre-trained visual features but omitting textual features. Song et al. [14] addressed these gaps with the personalized compatibility modeling scheme (GP-BPR), leveraging ResNet50 and personal preferences, demonstrating improved recommendations via multimodal features. Sagar et al. [5] further enhanced outfit compatibility, recommendation interpretability, and performance with PAI-BPR's attribute-wise modeling.

Several studies have improved fashion RSs by integrating category-specific relations and intra−/inter-modal compatibility among items [18–20]. Xu et al.'s [21] method captured style-compatible relationships through coarse- and fine-grained category views via multimodal integration, though it lacks personalization. Guan et al. [22] addressed this by introducing MG-PFCM, using metapaths and heterogeneous graphs to model complex user-item relationships.

Attention mechanisms enhance user preference modeling and item compatibility in RSs. Laenen et al. [23] showed their impact on item representation. ATRank [24] used it to capture user behaviors and relationships in heterogeneous data. Song et al. [17] used a co-attention mechanism to model complementary fashion items. Meta-path and co-attention mechanisms effectively capture complex user-item interactions [25,26] and improve recommendation tasks [27,28].

Our study builds upon the base model GP-BPR, which integrates visual and contextual data for fashion recommendations by merging item-item and user-item interactions. Although GP-BPR effectively balances outfit compatibility with personal preferences, it focuses mainly on visual features and lacks depth in personalization and user-item dynamics. We integrate meta-paths with a co-attention mechanism into the GP-BPR framework to address these limitations. Meta-paths models complex, multi-relational interactions across different modalities (e.g., visual, textual, contextual) [22], enhancing personalization underexplored in methods addressing only item compatibility [18–21]. We also refine the Metapath-Guided Personalized Fashion Compatibility Modeling (MG-PFCM) [22] to better capture item relationships. By introducing co-attention, our framework better-enhanced user and item embedding and nuanced relationships than MG-PFCM.

2.2. Choice overload and changes in consumer preference

Choice overload refers to a situation where choosing from many viable competing alternatives exceeds an individual's cognitive capacity [3]. It is associated with decreased preference, choice deferral, and product switching [2]. In outfit matching, multiple choices of tops can complement a specific bottom. A viable way to reduce choice overload is to employ RSs. However, user preferences can shift over time to a more coherent state before the final choice [11,29]. The existing literature has shown that preference shifting generally undergoes two phases: preference construction and preference stability [4]. However, the majority of outfit-matching recommendation studies [17,18,20,22] are operated under the assumption of stable preference by treating all interactions of the same user to be under the same user preference. This greatly ignores the fact that consumers frequently explore choices of outfits before moving towards a more coherent path of preferences [30]. Han et al. [31] introduced a Bi-LSTM model capturing temporal dependencies by treating clothing items as an ordered sequence, but it may overly rely on recent items, missing other important relationships. While some studies [15,29,31] acknowledge temporal factors, they don't fully examine how preferences evolve over time. Ding et al. [32] addressed this by modeling long-term preferences through user-item interactions and short-term preferences through item transitions. Yet, relying on others' outfit choices for short-term preferences may not align with the actual preferences of an individual. Additionally, consumers do not all come to a stable preference at the same rate. Therefore, assuming stable preferences can compromise the accuracy of RS recommendations. This study directly addresses this issue.

3. The proposed framework

Consider a set of users U, a set of tops T, and a set of bottoms B. Each user m has a history of preferred top-bottom pairs. The recommendation problem is then determining which bottom a user m would prefer to match a given top. Let $c_{i,j}^{m}$ represent the compatibility score (or preference) of user m for bottom j when paired with top i. Based on this score, a personalized ranking list of bottoms for a given top can be generated for personalized clothing matching.

To estimate $c_{i,j}^{m}$ effectively, we propose a compatibility modeling network F, which integrates the user's preference context to predict compatibility between fashion items as follows:

$$c_{i,j}^{m} = F(i,j,m|\Theta_F) \tag{1}$$

where Θ_F denotes the to-be-learned parameters of the model.

The proposed GPA-BPR for assessing the compatibility between tops and bottoms consists of two modules (see Fig. 1): the general outfit matching module, $\mathscr{G}(\bullet)$, and the personalized preference module, $\mathscr{P}(\bullet)$. By balancing the weights of these two modules, we can generate a compatibility score for bottom j when paired with top i. Formally, we have,

$$\left\{\begin{array}{c} c_{i,j}^{m} = \omega \bullet g_{i,j} + (1-\omega) \bullet p_{m,j}, \\ g_{i,j} = \mathscr{G}(i,j|\Theta_{\mathscr{G}}), \\ p_{m,j} = \mathscr{P}(m,j|\Theta_{\mathscr{P}}), \end{array}\right\} \tag{2}$$

where $\mathscr{G}(\bullet)$ and $\mathscr{P}(\bullet)$ correspond to the general outfit matching modeling and personalized preference modeling networks, respectively. $\Theta_{\mathscr{G}}$ and $\Theta_{\mathscr{P}}$ are the corresponding model parameters. $g_{i,j}$ represents the general compatibility calculated by the general outfit matching module $\mathscr{G}(\bullet)$, while $p_{m,j}$ denotes the compatibility score calculated by the personalized preference module $\mathscr{P}(\bullet)$. ω is a non-negative tradeoff parameter that controls the relative importance of the two modules.

In the following subsections, we first define the calculations for the general compatibility score $g_{i,j}$ and the personalized module compatibility score $p_{m,j}$. Next, we describe the BPR framework used to optimize the RS parameters Θ_F within a compatibility modeling network F for estimating the compatibility score $c_{i,j}^{m}$.

3.1. General outfit matching module

The general outfit matching module $\mathscr{G}(\bullet)$ calculates the compatibility score $g_{i,j}$ between tops and bottoms based on visual and textual features. Compatibility is represented by their distance in a latent space, capturing both feature types. This approach builds on GP-BPR [14], which has been shown to effectively model item compatibility and personal preferences.

Let v_i and v_j be the visual feature vectors and t_i and t_j the textual feature vectors of the top and bottom, respectively. The compatibility score $g_{i,j}$ between top i and bottom j is determined through matrix operations, where visual and textual features contribute proportionally based on a weighting factor ψ. The final compatibility score is obtained by summing the visual and textual components:

$$g_{i,j} = \psi(v_i)^T v_j + (1-\psi)(t_i)^T t_j \tag{3}$$

3.2. Personalized preference module

The personalized preference module $\mathscr{P}(\bullet)$ processes user IDs, item IDs, and user-item association paths to enhance personalized recommendations. It addresses gaps in implicit feedback left by previous models, focusing on recommending bottoms for a given top. The module consists of four steps: (1) generating user/bottom embeddings, (2) constructing meta-path-based embeddings, (3) refining embeddings through a co-attention mechanism, and (4) calculating the personalized compatibility score.

To generate user and bottom embeddings, we adopt the Neural Collaborative Filtering (NCF) framework [33] to generate user and bottom embeddings. User and bottom IDs are first transformed into one-hot encoded sparse vectors and then mapped to dense latent vectors via an embedding layer. These vectors, x_m for user m and y_j for bottom j, serve as latent feature representations. This approach allows for flexibility, as content features can be substituted for identity-based inputs in cases like cold-start issues.

Meta-path-based embeddings are derived through a Heterogeneous Information Network (HIN), where users, tops, and bottoms are nodes, and edges define relationships such as "user likes bottom" or "top pairs with bottom." Latent vectors for nodes are learned using Probabilistic Matrix Factorization (PMF) [34], which helps mitigate noise issues common in random walk methods [35]. Path instances are ranked by similarity, with low-similarity paths filtered out. We define meta-paths like U-T-B (user prefers bottom for a given top) and transform path instances into low-dimensional vectors using a CNN with max pooling. The final meta-path embedding r_ρ is obtained by averaging path instance vectors.

Since user preferences vary across different fashion attributes such as color, texture, brand, and style, a static embedding approach is insufficient. To refine user and bottom embeddings dynamically, we employ a co-attention mechanism inspired by [26]. This mechanism enhances personalized recommendations by adaptively weighting different meta-paths based on their relevance to specific users and items. Given embeddings x_m, y_j, and r_ρ for meta-path ρ, a two-layer structure is employed to generate the attention values:

$$\alpha_{m,j,\rho}^{(1)} = f\left(W_m^{(1)} x_m + W_j^{(1)} y_j + W_\rho^{(1)} r_\rho + b^{(1)}\right) \tag{4}$$

$$\alpha_{m,j,\rho}^{(2)} = f\left({w^{(2)}}^T \alpha_{m,j,\rho}^{(2)} + b^{(2)}\right) \tag{5}$$

where $f(\bullet)$ is set to the ReLU function. In the first layer structure, $W_*^{(1)}$ denotes the weight matrix, and $b^{(1)}$ represents the bias vector. In the second layer structure, ${w^{(2)}}^T$ represents the weight matrix, and $b^{(2)}$ represents the bias vector.

Let $P_{m\to j}$ denote a set of meta-paths that connect user m and bottom j. With multiple meta-paths, the final attention weight for meta-path ρ ($\rho \in P$) is calculated by normalizing all attention scores within P using the softmax function, as defined below:

$$\alpha_{m,j,\rho} = \frac{exp\left(\alpha_{m,j,\rho}^{(2)}\right)}{\sum_{\rho \in P_{m\to j}} exp\left(\alpha_{m,j,\rho}^{(2)}\right)} \tag{6}$$

After obtaining the meta-path attention scores $\alpha_{m,j,\rho}$, a new attention-based embedding for the interaction, denoted as $r_{m\to j}$, can be computed as:

$$r_{m\to j} = \sum_{\rho \in P_{m\to j}} \alpha_{m,j,\rho} \cdot r_\rho \tag{7}$$

With this refined interaction context, the attention vectors for user m and bottom j, denoted as β_m and β_j, are computed through a single-layer network:

$$\beta_m = f\left(W_m x_m + W_{m\to j} r_{m\to j} + b_m\right) \tag{8}$$

$$\beta_j = f\left(W_j' y_j + W_{m\to j}' r_{m\to j} + b_j'\right) \tag{9}$$

Where $f(\bullet)$ represents the ReLU function. W_* and b_m are the weight matrix and bias vector for the user attention layer, W_*' and b_j' are for the bottom attention layer.

The final user and bottom representations are computed by applying

Table 2
Dataset statistics.

Datasets	# of users	# of items	# of samples			
			Training	Validation	Testing	Total
IQON	3236	142,727	170,587	23,093	21,170	214,850
IQON-*ws*10γ01	3236	127,374	165,789	15,611	14,309	195,709
IQON-*ws*10γ005	3236	126,997	164,537	15,586	14,269	194,392
IQON-*ws*10γ001	3236	126,992	164,528	15,586	14,269	194,383
IQON-*ws*20γ01	3236	125,795	160,977	15,499	14,186	190,662
IQON-*ws*20γ005	3236	125,797	160,970	15,498	14,186	190,654
IQON-*ws*20γ001	3236	126,734	163,860	15,579	14,252	193,691
IQON-*ws*30γ01	3236	126,378	162,534	15,529	14,230	192,293
IQON-*ws*30γ005	3236	125,717	160,593	15,440	14,154	190,187
IQON-*ws*30γ001	3236	139,613	160,606	15,443	14,156	190,205

an element-wise product $\odot$ between the attention vectors β_m and β_j and the original embeddings. Through this attention mechanism, we obtain improved embeddings for the user, $\beta_m \odot x_m$, and the bottom, $\beta_j \odot y_j$.

To compute the personalized compatibility score, we combine traditional personalized preference modeling with meta-path-based scores. First, we apply MF to decompose the user-item preference matrix into lower-dimensional vectors, enabling accurate preference predictions. The personalized preference embedding is then constructed by concatenating the improved user embedding, improved item embedding, and the meta-path embedding. This composite representation is passed through an MLP with two hidden layers and ReLU activation function:

$$mp_{m,j} = \mathrm{MLP}\left(\beta_m \odot x_m \oplus \alpha_{m,j,\rho} \bullet r_\rho \oplus \beta_j \odot y_j\right) \tag{10}$$

Where $\oplus$ indicates the concatenation of the three embedding components.

The final personalized compatibility score of user m for bottom j is computed as:

$$p_{m,j} = d + e_m + e_j + \gamma_m^T \gamma_j + mp_{m,j} \tag{11}$$

Where d denotes the trainable global parameters, e_m and e_j are the bias terms, and γ_m and γ_j represent the user and item vectors derived from the decomposition of the preference matrix, with their inner product capturing the latent preference of user m for bottom j.

3.3. BPR recommendation algorithm

To accurately model interactions between users and bottoms, we adopt the BPR framework, known for its effectiveness in pairwise implicit preference modeling. We first construct a training set $D := \{(m,i,j,k) | m \in U \wedge (i,j) \in O_m \wedge k \in B \backslash j\}$, where O_m denotes the set of top-bottom pairs for user m and the quadruplet (m,i,j,k) represents that, given a top i, user m prefers the bottom j (positive sample) over k (negative sample) to create a proper outfit.

As mentioned in Section 3.1, to estimate the compatibility score $c_{i,j}^m$, the BPR framework trains a recommendation model by optimizing parameters Θ_F within a compatibility modeling network F. The objective is to find a set of Θ_F parameters that maximize the model's conditional probability. To enable gradient descent optimization, we use the BPR-Opt formula proposed in [6]. Consequently, the loss function in this study is defined as:

$$\mathscr{L} = \sum_{(m,i,j,k)\in\mathscr{D}} \left[-ln\left(\sigma\left(c_{i,j}^m - c_{i,k}^m\right)\right) \right] + \frac{\lambda}{2}\|\Theta_F\|_F^2 \tag{12}$$

4. Dataset

To evaluate our model's performance, we used the IQON3000 dataset [14], containing 308,747 outfits curated by 3568 users and featuring 672,335 fashion items. Sourced from the fashion platform IQON, each item includes an image and textual details such as name, ID, category, price, and brand. Tops and bottoms were classified into 39 and 34 types, respectively. Our study focuses on bottom-matching for a given top, tailored to individual users. To improve data quality, we cleaned IQON3000 by removing duplicate top-bottom pairs, yielding 216,773 unique outfits created by 3236 users and comprising 142,727 distinct fashion items. The dataset was split into training, validation, and test sets (8:1:1). Leave-one-out was used as cross-validation [14].

User preferences evolve over time, making it crucial to distinguish between preference construction and stability before generating outfit recommendations. During construction, users explore various clothing combinations across categories. As preferences stabilize, experimentation decreases, leading to more consistent choices. Identifying this shift enables recommendations based on stable user behavior. To measure preference stability, we filtered out early exploratory outfit sets, focusing on periods when preferences remained steady. More specifically, we tracked cumulative category counts for each user, monitoring distinct fashion categories (tops and bottoms) engaged with over time. This identifies consumer preference transitions for better recommendations.

Let TC be a set of top categories and BC be a set of bottom categories. Given a user m with a sequence of outfit category sets, defined as $S_m = \{(C_{i,1}, C_{j,1}), (C_{i,2}, C_{j,2}), \ldots, (C_{i,n}, C_{j,n})\}$, where each $(C_{i,q}, C_{j,q})$ represents an outfit category pair consisting of a top $C_{i,q} \in TC$ and a bottom $C_{j,q} \in BC$, and n denotes the total number of outfit sets uploaded by m. For a user- specified window size ws (where $1 \le ws \le n$), a total of $(n - ws + 1)$ subsequences can be derived from S_m. Let $S_{(m,q)} = \{(C_{i,q}, C_{j,q}), (C_{i,q+1}, C_{j,q+1}), \ldots, (C_{i,q+ws-1}, C_{j,q+ws-1})\}$ represent the q-th subsequence of S_m with window size ws. The cumulative number of outfit categories in $S_{(m,q)}$, denoted $N_{(m,q)}$, is calculated as the sum of the cumulative number of distinct top categories and the cumulative number of distinct bottom categories in $S_{(m,q)}$.

Given a user-specified threshold γ, we try to find the first two consecutive subsequences, i.e., $S_{(m,q)}$ and $S_{(m,q+1)}$, that satisfy the following criterion: $\frac{N_{(m,q+1)} - N_{(m,q)}}{N_{(m,q)}} < \gamma$. Then, they can be divided into two subsequences: preference construction sequence $\{(C_{i,1}, C_{j,1}), \ldots, (C_{i,q}, C_{j,q})\}$and preference stability sequence $S_m = \{(C_{i,q+1}, C_{j,q+1}), \ldots, (C_{i,n}, C_{j,n})\}$.

In our experiments, we set the window size ws to {10,20,30} and applied three threshold levels γ = {0.1, 0.05, 0.01} to quantify the rate of change in cumulative category numbers over time. These thresholds indicate how quickly user preferences stabilize, with slower changes marking the transition from exploratory to consistent preferences. Smaller window sizes detect short-term stabilization, suitable for users with frequent preference shifts, while larger window sizes capture long-term trends for gradual preference changes. Similarly, lower thresholds identify subtle transitions, whereas higher thresholds reflect quicker stabilization. Using these transition points, we refined the IQON dataset by removing outfit combinations from the preference construction phase, generating nine distinct datasets. This approach ensured consistent detection of stable phases across users. Table 2 presents dataset statistics, facilitating performance analysis under different stabilization assumptions and providing insights into preference evolution.

5. Experimental evaluation

5.1. Visual and textual embedding extraction

For textual embedding, we employed the Kuromoji Tokenizer for Japanese text segmentation, constructing a word index dictionary from the top 10,000 high-frequency words, with each word represented by its corresponding index. To ensure uniform input sequences, Zero Padding was applied. Subsequently, we employed the Nwjc2vec corpus, following the same strategy as [14], to generate 300-dimensional textual feature vectors, which were then further processed through a TextCNN model to obtain the final 400-dimensional textual feature vectors.

The five most popular pre-trained CNN models, including VGG16 [36], VGG19 [36], ResNet50 [37], InceptionV3 [38], and Inception-ResNetV2 [39], were used to extract latent space features from images. Each image was processed through a pre-trained network, with the final pooling layer's output providing a 2048-dimensional feature vector. Among them, InceptionV3 achieved the highest accuracy, scoring 0.7994 on the validation set and 0.7909 on the test set. Therefore, InceptionV3 was selected as the primary visual feature extraction method.

5.2. Experimental design

To answer the research questions outlined in Section 1, we designed three experiments using the IQON dataset and its preference stabilization phase subsets.

1. Baseline Comparison: To assess our model's effectiveness, we compared it against several baselines. POP measures item compatibility based on popularity in the training set, while RAND assigns random scores between positive and negative pairs. These are common lower-bound benchmarks in recommendation evaluations [5,13,14]. BPR-MF models latent user-item relationships through MF in a pairwise ranking framework [6], remaining a widely used method. VBPR [13] extends BPR-MF by integrating visual features, making it relevant for fashion recommendations. Its variants include TBPR [14], which replaces visual cues with textual descriptions, and VTBPR [14], which incorporates contextual factors for improved personalization. GP-BPR [14] models multi-modal user preferences using images and text, while PAI-BPR [5] enhances GP-BPR by incorporating fine-grained visual features, ranking among the best models for personalized outfit recommendations. In Experiment 1, we used two ranking-based metrics, AUC and MRR, to compare the baseline models with our GPA-BPR. AUC evaluates the model's ability to distinguish between relevant and irrelevant items, while MRR measures ranking quality by rewarding models that prioritize relevant items.
2. Stable Preference Modeling: To examine stable user preferences, we created preference stabilization phase datasets based on window size and threshold, resulting in nine variations (e.g., IQON-*ws*30γ001). These were compared against the baseline IQON dataset using both ranking-based metrics (AUC, MRR, and HitRate@10) and set-based metrics (Precision@10, Recall@10, and F1@10). HitRate@10 measures how often a relevant item appears in the top 10 recommendations. Precision@10 reflects the proportion of positive items in the top 10, while Recall@10 indicates the proportion of relevant items successfully recommended. F1@10 balances these two by computing their harmonic mean. Higher values across these metrics indicate better recommendation performance.

Table 3
Performance comparison among different approaches in terms of AUC and MRR.

Method	AUC	MRR
POP	0.6042	–
RAND	0.5014	–
BPR-MF [6]	0.7845	0.8451
VBPR [13]	0.7662	0.8090
TBPR [14]	0.8181	0.8433
VTBPR [14]	0.8171	0.8421
GP-BPR [14]	0.8424	0.7266
PAI-BPR [5]	0.8617	0.7829
GPA-BPR (our approach)	**0.8773**	**0.9016**

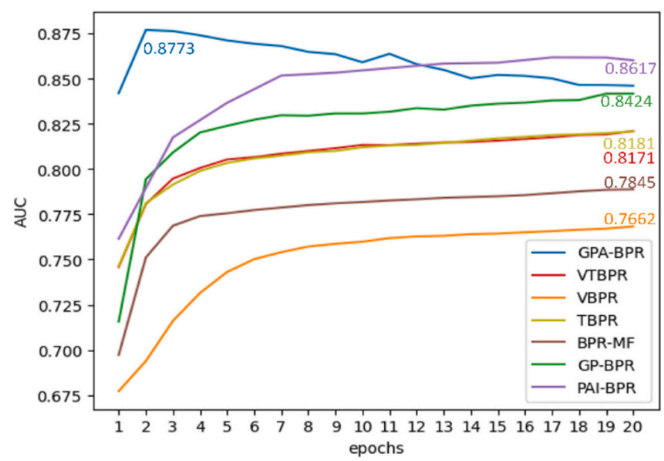

Fig. 2. The performance of the proposed model on AUC to each epoch.

3. Ablation Study: To validate the GPA-BPR model, we conducted ablation studies using both ranking-based and set-based metrics to assess the contributions of each model component. We tested two model variants: G-BPR, which excludes the personalized preference module, and PA-BPR, which excludes the general outfit matching module. By evaluating these variants individually, we determined the influence of each component on outfit matching, with the GPA-BPR model balancing both modules equally (weight of 0.5). This experiment addressed the framework's ability to generalize across varying personal preferences and product features by demonstrating each module's role in adapting outfit-matching recommendations.

All experiments utilized the adaptive moment estimation (Adam) optimizer to train all models with the PyTorch framework, tuning training parameters for each method independently on each dataset. The learning rate was set to 0.001, and the batch size was 256. The proposed model is fine-tuned for 20 epochs, and performance was evaluated on the test set.

5.3. Results

Table 3 compares the performance of baseline models with GPA-BPR, revealing key insights: (1) RAND, with random scoring (AUC = 0.5), serves as a basic benchmark, while POP, based on item popularity, better aligns with fashion trends. (2) VTBPR outperforms VBPR, TBPR, and BPR-MF, demonstrating the advantage of integrating image and text features. TBPR's higher AUC and MRR over VBPR suggest text features better capture user preferences. (3) GPA-BPR achieves the highest AUC (0.8773) and MRR (0.9016), surpassing PAI-BPR, GP-BPR, and other baselines, showing that combining outfit compatibility with personalized preferences, meta-path, and co-attention mechanisms enhances recommendation accuracy. Fig. 2 illustrates AUC trends over 20 epochs using the Adam optimizer (learning rate = 0.001), with GPA-BPR

Table 4
Performance comparison among IQON and different stable-phase datasets.

Datasets	AUC	MRR	HR@10	Pre@10	Rec@10	F1@10
IQON-$ws10\gamma01$	0.8876	0.9129	0.9234	0.2070	0.0314	0.0545
IQON-$ws10\gamma005$	0.8869	0.9056	0.9214	0.2108	0.0301	0.0527
IQON-$ws10\gamma001$	0.8844	0.9185	0.9264	0.2138	0.0305	0.0534
IQON-$ws20\gamma01$	0.8888	0.9087	0.9242	0.2123	0.0303	0.0530
IQON-$ws20\gamma005$	0.8838	0.9060	0.9184	0.2040	0.0291	0.0509
IQON-$ws20\gamma001$	0.8793	0.9123	0.9242	0.2149	0.0308	0.0539
IQON-$ws30\gamma01$	**0.8912**	0.9142	0.9251	0.2174	0.0316	0.0552
IQON-$ws30\gamma005$	0.8879	0.9124	0.9249	0.2145	0.0307	0.0537
IQON-$ws30\gamma001$	0.8897	**0.9215**	**0.9301**	0.2205	**0.0316**	**0.0553**
IQON	0.8773	0.9016	0.9171	0.2132	0.0214	0.0389
$\overline{\text{IQON} - ws30\gamma001}$	0.8615	0.8975	0.9134	**0.2439**	0.0280	0.0502

Note: HR = HitRate, Pre = Precision, Rec = Recall.

Table 5
Performance comparison on ablation analysis.

Datasets	Variants	AUC	MRR	HR@10	Pre@5	Rec@5	F1@5	Pre@10	Rec@10	F1@10
IQON	G-BPR	0.6519	0.2886	0.5305	0.2474	0.0124	0.0236	0.2998	0.0300	0.0545
	PA-BPR	0.8858	0.9394	0.9386	0.2861	0.0143	0.0272	0.2045	0.0205	0.0373
	GPA-BPR	0.8773	0.9016	0.9171	0.2895	0.0145	0.0276	0.2132	0.0214	0.0389
IQON -ws30γ001	G-BPR	0.6522	0.3265	0.6072	**0.3116**	**0.0157**	**0.0299**	**0.3850**	**0.0388**	**0.0705**
	PA-BPR	**0.8957**	**0.9446**	**0.9442**	0.2792	0.0141	0.0268	0.1961	0.0198	0.0360
	GPA-BPR	0.8897	0.9215	0.9301	0.3063	0.0155	0.0295	0.2205	0.0316	0.0554

Note: HR = HitRate, Pre = Precision, Rec = Recall.

peaking at epoch 2 before AUC declines, indicating meta-path and co-attention mechanisms accelerate convergence.

Table 4 compares the performance of preference stabilization phase datasets, showing they consistently outperform the baseline IQON dataset across multiple metrics. This indicates that focusing on stabilized user preferences leads to more accurate recommendations. IQON's AUC (0.8773) is significantly lower than all stabilization phase datasets, while its MRR (0.9016) is also surpassed by all stabilization phase datasets. HitRate@10 follows the same trend. Notably, IQON's F1@10 is the lowest (0.0389), whereas stabilization phase datasets achieve significantly higher scores. These improvements suggest the model not only aligns better with user preferences but also broadens relevant recommendations.

Among the preference stabilization phase datasets, IQON-$ws30\gamma001$ delivers the best results, with an AUC of 0.8897, MRR of 0.9215, and HitRate@10 of 0.9301. These metrics indicate that a larger window size with a lower threshold effectively captures stable preferences, enhancing recommendation accuracy. It also achieves a Recall@10 of 0.0316, the highest across stabilization datasets, balancing high precision and recall, resulting in the highest F1@10 score (0.0553).

Conversely, the preference construction phase subset $\overline{\text{IQON} - ws30\gamma001}$, representing early-stage preferences, performs significantly worse. Its AUC (0.8615) falls below both the baseline IQON dataset (0.8773) and all stabilization phase datasets, indicating reduced accuracy when construction-phase data is prioritized. Similarly, its MRR (0.8975) and HitRate@10 (0.9134) lag behind stabilization values, showing early-stage preference data is less useful for top recommendations. Interestingly, $\overline{\text{IQON} - ws30\gamma001}$ achieves higher Precision@10 (0.2439) than the baseline but lower Recall@10 (0.0280), reflecting a trade-off (i.e., greater precision but fewer relevant recommendations). Its F1@10 (0.0502) surpasses the baseline but remains below stabilization datasets, reinforcing the advantage of modeling stable-phase data.

Threshold analysis reveals that meaningful data segmentation occurs when γ ranges between 0.01 and 0.5. Values beyond this range, either above 0.5 or below 0.01, have negligible impact. $\gamma = 0.01$ effectively captures subtle preference changes, while $\gamma = 0.5$ defines the upper limit for rapidly stabilizing users. Additionally, window size plays a critical role, with $ws = 30$ consistently yielding the best MRR and AUC, confirming that larger windows better capture long-term preferences. These findings establish $\gamma = 0.01$ and $ws = 30$ as the optimal combination for detecting stabilized preferences and enhancing recommendations.

The performance comparison in Table 5 across different model variants shows that GPA-BPR is the most robust and well-balanced model overall, delivering strong results consistently across both datasets and a wide range of evaluation metrics. In the IQON dataset, PA-BPR achieves the highest MRR (0.9394) and HitRate@10 (0.9386), indicating that purely personalized signals are effective in capturing user-specific ranking patterns. However, this comes at a cost, as PA-BPR underperforms in set-level metrics such as Precision@5 (0.2861), Recall@5 (0.0143), and F1@5 (0.0272), as well as in Precision@10, Recall@10, and F1@10, suggesting that it fails to generate cohesive outfit recommendations. G-BPR, by contrast, models only the general outfit compatibility and performs better than PA-BPR in most set-level metrics, but its overall ranking performance is poor. For example, its MRR drops to 0.2886, a substantial decline compared to GPA-BPR's 0.9016. In the more challenging IQON-$ws30\gamma001$ dataset, these trade-offs become even more evident. PA-BPR again leads in MRR (0.9446) but shows the lowest Recall@5 (0.0141) and F1@5 (0.0268), while G-BPR achieves the highest Precision@10 and F1@10, yet still suffers from low MRR (0.3265). GPA-BPR effectively balances these extremes, ranking consistently high across both ranking-oriented and set-level metrics, including strong performance in Precision@5 (0.3063), Recall@5 (0.0155), and F1@5 (0.0295). This demonstrates its ability to generate recommendations that are not only personalized but also cohesive and contextually appropriate as outfits. Moreover, its stable performance across datasets of varying difficulty highlights GPA-BPR's robustness and adaptability. Unlike G-BPR and PA-BPR, which exhibit strengths under specific conditions, GPA-BPR maintains reliable performance across all key scenarios, making it the most versatile and dependable model overall.

6. Discussion

6.1. Recommendation performance analysis of GPA-BPR

Our results affirm the superior performance of the proposed GPA-

Fig. 3. The top 10 ranking list for VBPR, GP-BPR, and our method.

BPR model compared to multiple baseline approaches, including those that integrate multimodal information such as VBPR, TBPR, and GP-BPR. The results in Table 3 indicate that GPA-BPR outperforms all related models, including GP-BPR, which combines an outfit compatibility module with a personalized preference module. This underscores the strength of our approach in incorporating meta-path and co-attention mechanisms, which effectively capture deeper implicit information and personalized judgments. These enhancements result in a better alignment with user preferences and outfit compatibility.

Additionally, as seen in Fig. 2, GPA-BPR achieves its peak AUC at epoch 2, with further training leading to a slight decline. This rapid convergence suggests that the integration of meta-path and co-attention enhances model accuracy, reducing the need for extensive training epochs.

The ablation studies in Table 5 further illuminate the contributions of different model components to recommendation performance. Notably, G-BPR, which excludes the personalized preference module, shows the lowest performance across metrics, especially in AUC, highlighting the indispensability of capturing consumer preferences. PA-BPR (w/o G) achieves a significant improvement in AUC (0.8858), indicating that personalization can bring notable benefits. However, GPA-BPR reaches the AUC at 0.8773, demonstrating that combining personalized preference with meta-path and co-attention mechanisms achieves the most balanced and effective results. These findings collectively suggest that GPA-BPR's success is rooted in its layered approach in that meta-path and co-attention allowed the model to capture nuanced patterns of user behavior and item compatibility, while the preference module ensured alignment with individual user tastes.

6.2. Case study

To evaluate the effectiveness of our approach, we conduct a case study comparing the ranking results of our GPA-BPR model with two baseline models, VBPR and GP-BPR. While all three models share the same foundational framework, they incorporate distinct features: VBPR integrates image-based features within BPR, GP-BPR incorporates textual features, and GPA-BPR further enhances these by leveraging meta-path information and a co-attention mechanism. We randomly selected a user (ID 1468) with queries for tops of varying styles and colors, and each model generated the top 10 recommended bottoms from a pool of 20 items.

As shown in Fig. 3, a white top was chosen as the query item; the user's historical data indicates a preference for casual bottoms, including shorts, wide straight-leg trousers, and straight-leg jeans, as well as skirts like mid-length straight skirts, A-line long skirts, and knee-length pleated skirts. The recommendation results show all three models successfully identify a relevant bottom item, but the rankings vary, revealing differences in each model's approach to user preferences and item selection. VBPR ranks items based on image aesthetics and recommends some casual bottoms; however, it lacks alignment with the user's tastes, placing irrelevant items like red corduroy shorts and narrow trousers in positions 9 and 10. GP-BPR improves upon VBPR by filtering out irrelevant items and including casual shorts and skirts aligned with the user's preferences, but it ranks tapered trousers second. GPA-BPR balances historical preferences with item compatibility, placing casual shorts and mid-length skirts higher while positioning casual trousers further down. In the top 10 recommendations, GPA-BPR's selection (IQON-*ws*30γ001, which reflects stable preferences) shows diversity, with both pants and skirts aligned with the user's historical style.

6.3. Empirical validation

Multiple tops could complement the same bottom in outfit matching, which potentially causes choice overload. We adapted the established instrument of decision efficiency and satisfaction from [40,41], respectively. The questionnaire was administered to two groups of randomly selected respondents: one that went through the recommendations by the RS (the Recomm group) and the other one that did not (the No_Recomm group). Each group had 125 responses. A multi-group comparison with bootstrapping using SmartPLS was performed to study the relationship between Perceived Decision Efficiency (PDE) and Satisfaction (SAT). The path coefficient of PDE→SAT was higher for the Recomm group (No_Recomm = 0.634 vs. Recomm = 0.853) ($p = 0.048$). The Recomm group also had a higher R^2 value (No_Recomm = 0.616 vs. Recomm = 0.792). These results show that decision efficiency (PDE) provided by the RS had a stronger effect on consumer satisfaction (SAT) for the Recomm group than the No_Recomm group. Additionally, more variability of consumer satisfaction was explained by decision efficiency for the Recomm group.

Table 6
Contributions of the proposed framework.

Research challenge	Proposed method	Experimental evidence
Simplified user-item interactions. [7,15,18,23]	Personalized preference module using meta-path and co-attention.	GPA-BPR achieves an AUC of 0.8773, surpassing baselines.
Limited personalization and multimodal integration. [7,13,16,20,21]	Incorporation of textual, visual, and meta-path embeddings.	GPA-BPR achieves an AUC of 0.8897 and MRR of 0.9215, demonstrating improved outfit compatibility.
Assumption of static user preferences. [2,5,22,24,32]	Segmentation into construction and stabilization phases.	Stabilization datasets outperform IQON across all parameters, ensuring robust recommendations.

6.4. Framework evaluation and contributions

Despite advancements in RS studies, they continue to encounter key challenges. Models like VBPR [13] and Deep Visual BPR [16] integrate visual features but lack personalization and fail to fully utilize multimodal data. GP-BPR [14] incorporates contextual and visual features but cannot capture fine-grained, multi-relational user-item interactions. Most RS studies overlook the transition between preference construction and stabilization. These limitations highlight three core challenges: (1) oversimplified user-item interaction modeling, (2) insufficient personalization and multimodal integration, and (3) the static assumption of user preferences.

Table 6 outlines our key contributions. By integrating personalized preference modeling, multimodality, and preference phase segmentation, GPA-BPR better captures user-item interactions and changes in user preferences, resulting in better recommendation accuracy.

7. Conclusion

This research advances personalized outfit matching by predicting suitable fashion items for users. We propose GPA-BPR, a compatibility modeling framework integrating images, textual descriptions, and meta-path data for personalized recommendations. GPA-BPR leverages InceptionV3 for image features, TextCNN for textual features, and collaborative filtering MF for latent preferences. A co-attention mechanism captures user-item-meta-path interactions, enhancing personalization and recommendation performance. This approach effectively addresses both fashion compatibility and user preferences.

Our study makes several key theoretical contributions. First, it is the first to extract latent associations between users and fashion items for personalized outfit matching. By leveraging a co-attention mechanism, we capture users' evolving interests across items, refining personalized preferences while mitigating choice overload. Second, we employ a rigorous feature extraction methodology, providing a valuable reference for future research in this domain. Third, most outfit-matching RSs treat all interactions as stable consumer preferences, but research shows otherwise [11]. To address this gap, we develop a strategy to distinguish between preference construction and stabilization, demonstrating superior performance over existing approaches. This study offers several practical implications. The proposed framework can be applied to e-commerce platforms, improving outfit recommendations and potentially increasing sales. The required input data, such as item images, textual descriptions, and user interaction history, is readily available, facilitating seamless implementation. While our study focuses on tops and bottoms, future research can extend this approach to accessories, footwear, and complete outfit recommendations.

CRediT authorship contribution statement

Ya-Han Hu: Writing – review & editing, Visualization, Project administration, Methodology, Funding acquisition, Formal analysis. **Ting-Hsuan Liu:** Writing – original draft, Resources, Formal analysis, Data curation. **Kuanchin Chen:** Writing – review & editing, Validation, Conceptualization. **Fan-Chi Yeh:** Writing – original draft, Validation, Software, Data curation.

Declaration of competing interest

The authors report no declarations of interest.

Data availability

Data will be made available on request.

Acknowledgements

This research was supported in part by the National Science and Technology Council (grant numbers MOST 111-2410-H-008-026-MY2 and NSTC 113-2410-H-008-054).

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